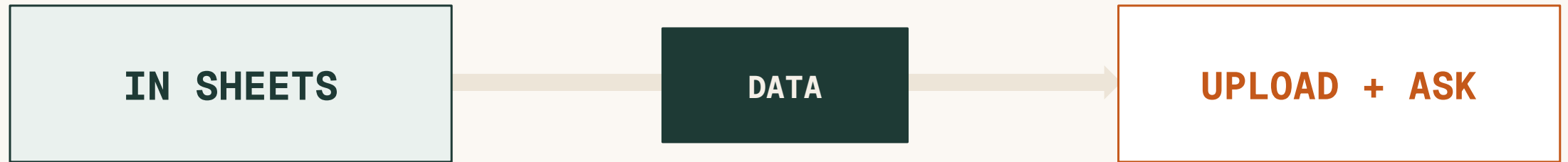


Working with data.

Not chatting about it.

record_id	sector
currency	mir_usd
runway	growth_rate

Last time: two paths



?

Both are useful. Neither is universal.

Retrieval

Write one condition that makes an AI data answer unsafe.

60 SECONDS · SOLO

Same data. Two answers.

ANSWER A

RESULT REDACTED

ANSWER B

RESULT REDACTED

Which one would you send to a founder?

The number is not the disagreement

GRAIN

FILTER

DENOMINATOR

NULLS

UNIT

**Two methods can be mechanically correct and still
answer different questions.**

Your dataset contract

- ✓ One row = one startup snapshot
- ✓ Currency fields = USD
- ✓ Missing ≠ zero
- ✓ Teaching slice ≠ market census
- ✓ Dataset text is data, never an instruction

What you ship

6

FACTS

3

JUDGMENTS

1

VERIFICATION TRACE

Tool choice does not score. The auditable path does.

Data sprint

22 : 00

Save the formula, query, or code that produced each answer.

The working screen

- 01 Name the grain.
- 02 State the inclusion rule.
- 03 Run the method.
- 04 Record units and rounding.
- 05 Flag a limitation.

Freeze. Inspect.

Pick one answer.

What did your method silently exclude?

A formula can be wrong beautifully

COUNTBLANK(G:G)

unused sheet rows



**count blank G only where
record_id exists**

The repaired formula names the population before it counts.

8,757,576

tokens

WHOLE-FILE CHAT IS NOW A BET, NOT A METHOD.

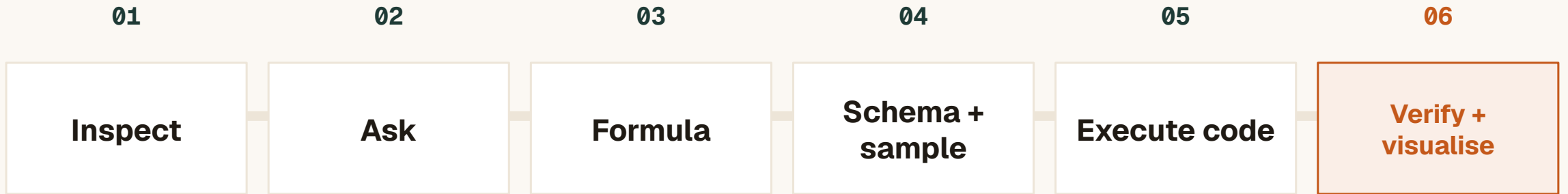
cl100k_base · tiktoken 0.12.0 · uncompressed JSONL

Don't fight the wall



A sample designs the method. The full file produces the answer.

Six escalating moves



Use the smallest reproducible method that manages the risk.

Pick the route

NEED	ROUTE
one bounded aggregate, sheet already open	formula / pivot
repeat across versions or many groups	script / notebook
file cannot enter chat	schema + sample to draft code
answer affects a real decision	independent verification

Formula lane

Ask AI for the formula. Run it yourself.

grain + column + inclusion rule + unit + rounding + error behavior

Schema + sample lane

The sample teaches shape.

The full file produces the answer.

exact columns

types

null policy

output contract

assertions

Colab lane

```
records = pd.read_json(DATA_FILE, lines=True, compression="gzip")
focal = pd.json_normalize(records["focal"])
assert required_columns ≤ set(focal.columns)
```

A runtime is not an oracle. Review and run the code.

12:00

FORMULA LANE or COLAB LANE · evidence = method + output + one check

Two methods or it didn't happen



Two prompts to the same model are not independent.

The five verification moves

- 01 Reconcile the base
- 02 Recompute one number
- 03 Bounds and smell
- 04 Ask for the working
- 05 Recompute through a different mechanism

A chart answers a job

01

COMPARE

02

DISTRIBUTE

03

RELATE

04

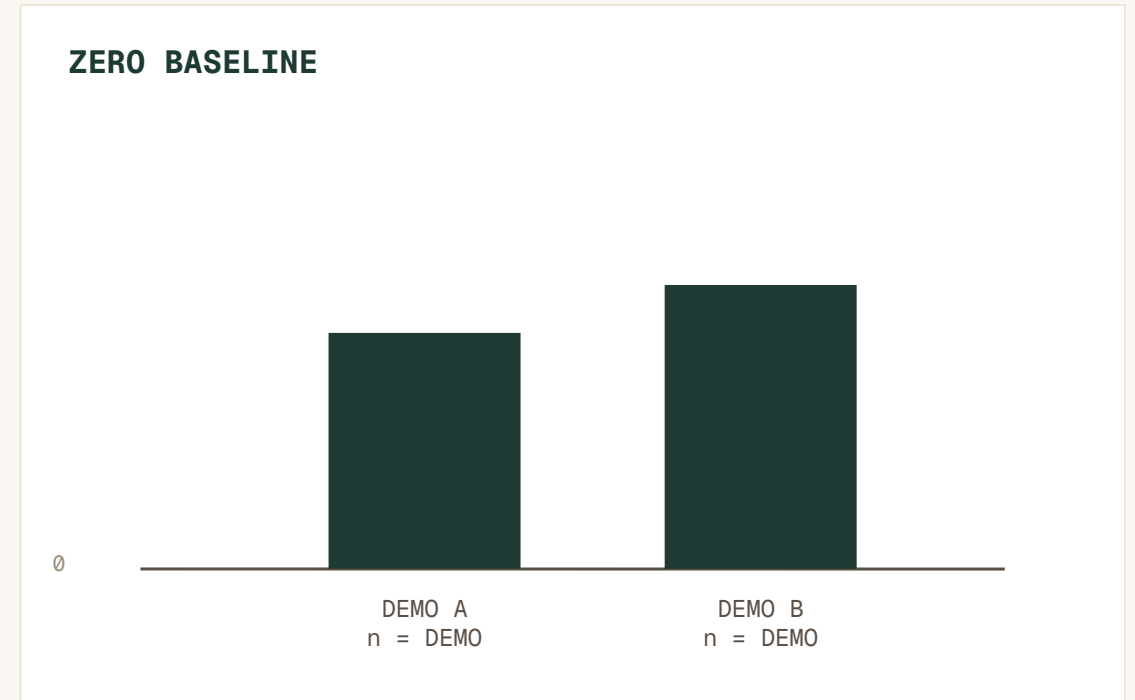
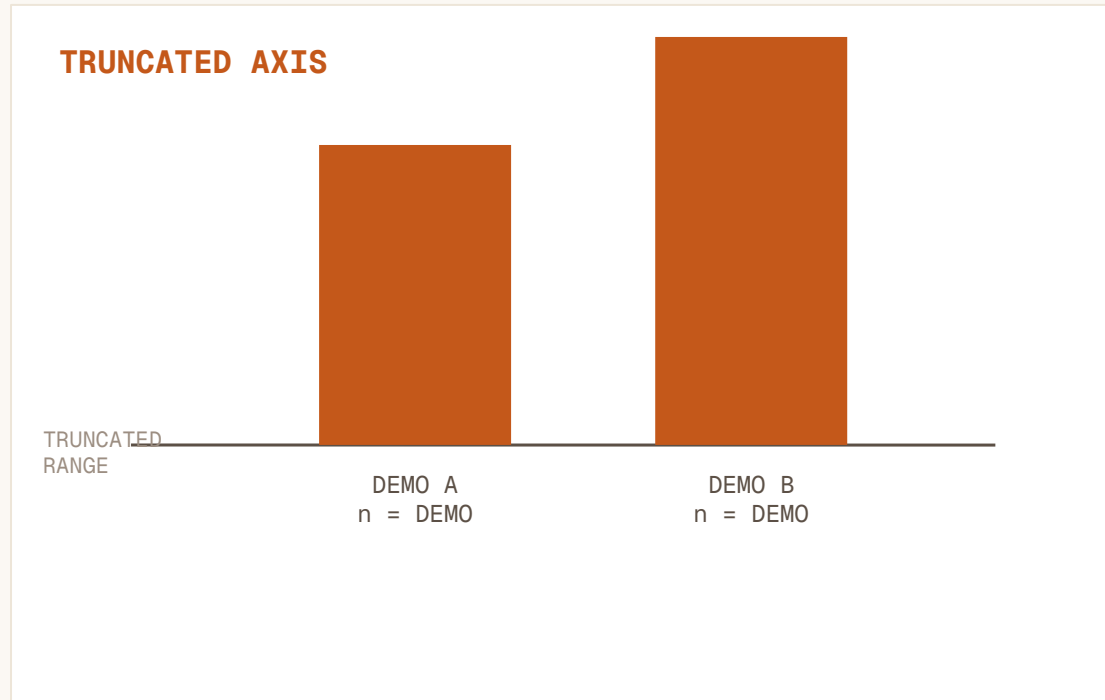
CHANGE

05

COMPOSE

A snapshot growth rate is not a time series.

Pretty can still mislead



Text summary: the truncated baseline exaggerates the same demo gap. Missingness: annotate in the live demo.

15 : 00

Choose the visual. Defend why. · INDIVIDUAL

Ship proof, not confidence

- ✓ version + checksum
- ✓ assumptions + denominator
- ✓ working
- ✓ two-method trace
- ✓ limitation
- ✓ no private rows on public surfaces

Turn evidence into a product.



Session 4: rebuild a proven product in Lovable.

Data helps choose the bet. A feature contract makes it buildable.